

Commerce XI Maths App

Miscellaneous Exercise 3

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 3 - Skewness | Total questions: 12

Question 1 - ENTER ANSWER · PG-44 · MEX-Q1

For a distribution, mean = 100, mode = 80 and S.D. = 20.
Find Pearsonian coefficient of skewness Sk_p .

Correct answer: 1

Explanation and steps

$$Sk_p = \frac{100 - 80}{20} = 1.$$

Question 2 - ENTER ANSWER · PG-44 · MEX-Q2

For a distribution, mean = 60, median = 75 and variance = 900.
Find Pearsonian coefficient of skewness Sk_p .

Correct answer: -1.5

Explanation and steps

$$S.D. = 30. Sk_p = 3 \left(\frac{60 - 75}{30} \right) = -1.5.$$

Question 3 - ENTER ANSWER · PG-44 · MEX-Q3

For a distribution, $Q_1 = 25$, $Q_2 = 35$ and $Q_3 = 50$.
Find Bowley's coefficient of skewness Sk_b .

Correct answer: 0.2

Explanation and steps

$$Sk_b = \frac{50 + 25 - 70}{50 - 25} = \frac{5}{25} = 0.2.$$

Question 4 - ENTER ANSWER · PG-44 · MEX-Q4

For a distribution, $Q_3 - Q_2 = 40$ and $Q_2 - Q_1 = 60$.
Find Bowley's coefficient of skewness Sk_b .

Correct answer: -0.2

Explanation and steps

$$Sk_b = \frac{40 - 60}{40 + 60} = \frac{-20}{100} = -0.2.$$

Question 5 - ENTER ANSWER · PG-44 · MEX-Q5

For a distribution, $Sk_\beta = 0.6$, $Q_2 = 38$ and $Q_1 + Q_3 = 100$.
Find Q_3 .

Correct answer: 70

Explanation and steps

$$0.6 = \frac{100 - 76}{Q_3 - Q_1}, \text{ so } Q_3 - Q_1 = 40. \text{ Solving with } Q_1 + Q_3 = 100 \text{ gives } Q_3 = 70.$$

Question 6 - ENTER ANSWER · PG-44 · MEX-Q5

For a distribution, $Sk_\beta = 0.6$, $Q_2 = 38$ and $Q_1 + Q_3 = 100$.
Find Q_1 .

Correct answer: 30

Explanation and steps

From $Q_3 - Q_1 = 40$ and $Q_1 + Q_3 = 100$, $Q_1 = 30$.

Question 7 - ENTER ANSWER · PG-44 · MEX-Q6

For a frequency distribution, the mean is 200,
the coefficient of variation is 8% and Karl Pearsonian's coefficient of skewness is 0.3.
Find the mode of the distribution.

Correct answer: 195.2

Explanation and steps

$$\text{S.D.} = 8\% \text{ of } 200 = 16. \text{ Hence } 0.3 = \frac{200 - \text{Mode}}{16} \text{ and Mode} = 195.2.$$

Question 8 - ENTER ANSWER · PG-44 · MEX-Q6

For a frequency distribution, the mean is 200,
the coefficient of variation is 8% and Karl Pearsonian's coefficient of skewness is 0.3.
Find the median of the distribution.

Correct answer: 198.4

Explanation and steps

$$\text{Mean} - \text{Mode} = 4.8 = 3 \times (\text{Mean} - \text{Median}), \text{ so Median} = 198.4.$$

Question 9 - ENTER ANSWER · PG-44 · MEX-Q7

Marks above	0	10	20	30	40	50	60	70	80
No. of students	120	115	108	98	85	60	18	5	0

Calculate Karl Pearsonian coefficient of skewness Sk_p .

Correct answer: -0.4760

Explanation and steps

Convert the more-than cumulative frequencies to class frequencies, then compute mean, mode and S.D.; $Sk_p \approx -0.4760$.

Question 10 - ENTER ANSWER · PG-44 · MEX-Q8

Marks above	0	10	20	30	40	50	60	70	80
No. of students	120	115	108	98	85	60	18	5	0

Calculate Bowley's coefficient of skewness Sk_β .

Correct answer: -0.3194

Explanation and steps

Find Q_1 , Q_2 and Q_3 from the converted grouped distribution; $Sk_\beta \approx -0.3194$.

Question 11 - ENTER ANSWER · PG-44 · MEX-Q9

Find Sk_p for the observations:

18, 27, 10, 25, 31, 13, 28

Correct answer: -1.3182

Explanation and steps

Using the empirical median form because the mode is indeterminate gives $Sk_p \approx -1.3182$.

Question 12 - ENTER ANSWER · PG-44 · MEX-Q10

Find Sk_β for the observations:

18, 27, 10, 25, 31, 13, 28

Correct answer: -0.6

Explanation and steps

Ordered data: 10, 13, 18, 25, 27, 28, 31. Thus $Q_1 = 13$, $Q_2 = 25$, $Q_3 = 28$ and $Sk_\beta = \frac{28 + 13 - 50}{15} = -0.6$.